

Google Play Store App Insights Dashboard – Case Study Report (Looker Studio)

1. Executive Summary

The Google Play Store is one of the largest mobile application marketplaces, containing millions of apps across multiple categories. This project aims to analyze app performance, user engagement, pricing strategies, app popularity, and update trends using the Google Play Store dataset.

A five-page interactive dashboard was developed in Looker Studio to transform raw data into meaningful business insights. The dashboard helps stakeholders understand market trends, user preferences, monetization opportunities, and factors influencing app success.

2. Business Objective

The primary objectives of this project are:

- Analyze the distribution of applications across categories.
 - Evaluate user satisfaction through ratings and reviews.
 - Identify the most popular apps and categories based on installs.
 - Understand pricing strategies of free and paid applications.
 - Examine the impact of app size on popularity and ratings.
 - Study update trends and Android version compatibility.
 - Generate actionable recommendations for app developers and businesses.
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3. Dataset Overview

The dataset contains information about applications available on the Google Play Store.

Key Attributes

- App Name
- Category
- Rating
- Reviews
- Installs
- Type (Free/Paid)
- Price
- Content Rating
- Size
- Android Version
- Last Updated

Dataset Statistics

Metric	Value
Total Apps	8,194
Categories	Multiple
Free Apps	7,590
Paid Apps	604
Average Rating	4.17
Total Installs	75 Billion+
Total Reviews	2.1 Billion+

4. Data Cleaning & Preparation

Data quality plays a crucial role in analytics. Several preprocessing steps were performed before dashboard creation.

Data Cleaning Steps

1. Missing Values Handling

- Removed irrelevant missing records.
- Retained valid null values where appropriate.

2. Size Conversion

- Converted KB and MB values into a unified Size_MB column.

3. Installs Cleaning

- Removed commas and plus signs.
- Converted install counts into numeric values.

4. Date Transformation

- Converted Last Updated field into date format.
- Extracted Update Year for trend analysis.

5. Feature Engineering

Created:

- Rating Group
- Rating Sort
- Install Group
- Install Group Sort
- Price Group
- Update Year

These fields improved dashboard readability and analysis.

5. Dashboard Page 1: Executive Overview

The Executive Overview page provides a high-level summary of the Google Play Store ecosystem.

KPIs

- Total Apps: 8,194
- Total Installs: 75 Billion+
- Average Rating: 4.17
- Total Reviews: 2.1 Billion+
- Free Apps: 7,590
- Paid Apps: 604

Key Visualizations

Top Categories by Number of Apps

The Family category contains the largest number of applications, followed by Game and Tools.

Free vs Paid Apps

Free applications dominate the marketplace, accounting for approximately 92.6% of all apps.

Content Rating Distribution

Most applications target general audiences through the "Everyone" content rating.

App Updates Over Time

A significant increase in updates is observed in 2018, indicating active app maintenance.

Business Insights

- Free apps dominate the market.
 - Family and Game categories contain the highest number of apps.
 - Most applications target broad audiences.
 - Developers increasingly focus on continuous updates.
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6. Dashboard Page 2: Ratings & User Satisfaction Analysis

This page evaluates user satisfaction using ratings and reviews.

KPIs

- Average Rating: 4.17
- Apps Rated 4+: 6,300+
- Maximum Rating: 5
- Minimum Rating: 1

Key Findings

Rating Distribution

Most applications fall within the 4–5 rating range.

Top Categories by Average Rating

Categories such as Events, Education, and Art & Design demonstrate the highest average ratings.

Rating vs Reviews

Applications with large review counts generally maintain strong ratings, indicating good user experience.

Business Insights

- Most apps receive favorable ratings.
 - High ratings contribute to user trust.
 - User satisfaction remains strong across major categories.
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7. Dashboard Page 3: Install & Popularity Analysis

This page focuses on app popularity and market demand.

KPIs

- Total Installs
- Average Installs per App
- Apps Above 1 Million Installs

Key Findings

Most Installed Apps

A small number of apps contribute significantly to total installs.

Top Categories by Installs

Game, Communication, and Social categories attract the highest install volumes.

Install Distribution

Most applications belong to lower install brackets, while a small percentage achieve extremely high adoption.

Installs vs Ratings

Popular apps generally maintain good ratings, suggesting a positive relationship between quality and adoption.

Business Insights

- Install concentration is highly uneven.
 - User demand is concentrated within a few categories.
 - Quality and popularity often go hand-in-hand.
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8. Dashboard Page 4: Pricing & Revenue Analysis

This page investigates monetization strategies and pricing trends.

KPIs

- Free Apps: 7,590
- Paid Apps: 604
- Average Price
- Maximum Price: \$400

Key Findings

Free vs Paid Apps

Free applications represent more than 90% of the marketplace.

Paid Apps by Category

Family, Game, and Personalization categories contain the highest number of paid apps.

Price Distribution

Most paid apps are priced within lower price ranges.

Price vs Rating

Higher prices do not guarantee higher ratings.

Price vs Reviews

Expensive applications often receive fewer reviews than free apps.

Business Insights

- Freemium remains the dominant business model.
 - Affordable pricing increases accessibility.
 - User experience matters more than price.
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9. Dashboard Page 5: App Size & Update Trends Analysis

This page evaluates app optimization and maintenance patterns.

KPIs

- Average App Size: 21.76 MB
- Largest App: 100 MB
- Apps Updated in 2018: 270
- Latest Update Year: 2018

Key Findings

App Size Distribution

Most applications fall within moderate size ranges.

Average App Size by Category

Entertainment apps are generally larger than apps in other categories.

Size vs Installs

Larger app size does not guarantee more installs.

Size vs Ratings

App size has minimal impact on user ratings.

Update Trends

2018 witnessed the highest update activity.

Android Version Distribution

Most applications support modern Android versions.

Business Insights

- App optimization remains important.
 - Frequent updates improve user confidence.
 - Compatibility affects market reach.
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10. Key Findings

Major Discoveries

1. Free apps dominate the Play Store ecosystem.
 2. Average app ratings are generally high.
 3. Family and Game categories lead in app count.
 4. Popular categories generate the highest install volumes.
 5. Pricing has limited influence on ratings.
 6. User experience drives app success.
 7. App size does not significantly affect popularity.
 8. Continuous updates indicate active development.
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11. Business Recommendations

For Developers

- Focus on user experience and app quality.
- Maintain ratings above 4.0.
- Optimize app size for better performance.
- Release updates regularly.
- Support multiple Android versions.

For Businesses

- Prioritize high-demand categories.
 - Adopt freemium monetization models.
 - Use customer feedback to improve products.
 - Monitor ratings and reviews continuously.
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12. Conclusion

The Google Play Store App Insights Dashboard successfully transformed raw application data into meaningful business intelligence. The analysis revealed that free apps dominate the marketplace, user ratings remain consistently high, install volumes are concentrated among a small set of applications, and continuous updates are critical for long-term success.

The dashboard provides stakeholders with a comprehensive view of app performance, helping developers and businesses make informed decisions regarding product development, pricing strategies, user engagement, and market positioning. This project demonstrates the practical application of data analytics and business intelligence tools in solving real-world business problems.